It Takes More Than Seeds

Knowledge, Capital, and Women's Agency in Nigeria

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Background & Country Context

- **Nigeria's agricultrual sector:**

24% of GDP	**80% are smallholder farmers**	**80% of HHs in north 50% of HHs in south**	**36.2% Women**

- **Gender divide within agricultural sector in Nigeria**
  - The World Bank estimates that female plot managers in Nigeria produce roughly 30% less per hectare than male farmers and generate 25% less output per hectare after controlling for observable household characteristics
  - The aggregate cost of this difference is estimated at $2.3 billion annually, or about 0.6 percent of Nigeria's GDP, with broader spillover gains of up to 2 percent of GDP if the productivity gap were closed (World Bank, 2023).
- **Nigeria's unique context within the African continent**
  - The gender gap in agricultural labor productivity in Nigeria is large relative to other sub-Saharan African contexts (Palacios-López, 2015) and it cannot be explained fully by observable differences in endowments (Mukasa, 2015).
  - Female-managed plots in N. Nigeria are 42% smaller than male-managed plots.
  - Women rely more heavily on female household labor per hectare and the gender productivity gap is wider in the north than in the south (Oseni, 2015).

Motivation for Intervention

What is our focus?

THE PROBLEM

Unexplained gap

77% of the productivity difference between male and female farmers is attributable to unexplained structural factors rather than to differences in endowments (Mukasa, 2015).

interventions that target only the observable aspects of the gap, such as inputs or credit, are unlikely to generate durable gains if the underlying institutional and social mechanisms remain unchanged.

THE CONSEQUENCE

Reinforcement of gap

Productive resources such as land, credit, inputs, and knowledge are more often directed toward men than toward women, reinforcing the perception that men are the farm's legitimate economic agents while women are dependents rather than contributors (Oseni, 2015).

The identity of the resource recipient shapes productive outcomes in ways that aggregate quantities of inputs alone cannot capture.

THE NUANCE

Participation != bargaining power

Women's increased participation in agricultural value chains does not necessarily improve their control over the resulting income, and male household members may capture commercially successful female managed activities (Swaans, 2015).

A primary reason many interventions fail is their exclusive focus on visible outputs rather than the structural drivers that sustain them (Farnworth, 2015).

We study whether directing agricultural extension services AND microcredit exclusively to women increases women's decision-making authority and control over farm resources, and whether these shifts alter their perceived role as productive contributors within the household.

Intervention Logic

Theory of Change

THE PROBLEM

Resources flow to men

Extension, credit, and inputs go to male household heads — even when women do most of the farm labor.

77% of Nigeria's productivity gap cannot be explained by observable differences alone (Mukasa & Salami, 2015).

→

THE MECHANISM

Institutional Signaling

When institutions name women as recipients of knowledge and capital, they shift the household perception of women as legitimate economic actors — not dependents.

→

THE INTERVENTION

T1 + T2

T1 Women-only extension
T2 Extension + microcredit

Both arms make women the publicly named recipient of resources.

What We're Testing

Three-Arm RCT · Village-level randomization

CONTROL	TREATMENT 1	TREATMENT 2
Status Quo	**Women-Only Extension**	**Extension + Microcredit**
No study-provided services. Counterfactual: what happens without gender-targeted resources?	6 modules over one growing season. Female facilitators. Woman is the named recipient of knowledge.	All of T1 plus women are the sole eligible borrowers, disbursed in the woman's name via mobile money.

T1 vs Control → effect of knowledge alone **T2 vs Control** → joint effect of knowledge + capital **T2 vs T1** → marginal effect of credit

Primary Outcomes

↑ Women's decision-making authority ↑ Control over farm revenue ↑ Perceived productive role

Study Design: Sample

- **Randomization at the village level:**
  - Mechanism operates through community-level visibility
  - Minimizes risk of contamination between groups as extension knowledge diffuses across households
- **Stratified Sampling by Region:**
  - Rural farming villages selected from northern and southern states
    - To capture gender norm intensity: northen Nigeria is predominantly Muslim and Hausa-Fulani households where Purdah is followed. We use this as proxy for norm-intensity where social restrictions on women are strongest.
  - Eligible households: at least one adult woman (aged 15 or more) engaged in agricultural production
    - Village-level baseline FLFP in agriculture used to identify villages
- **Sample size:**

135 Total Villages

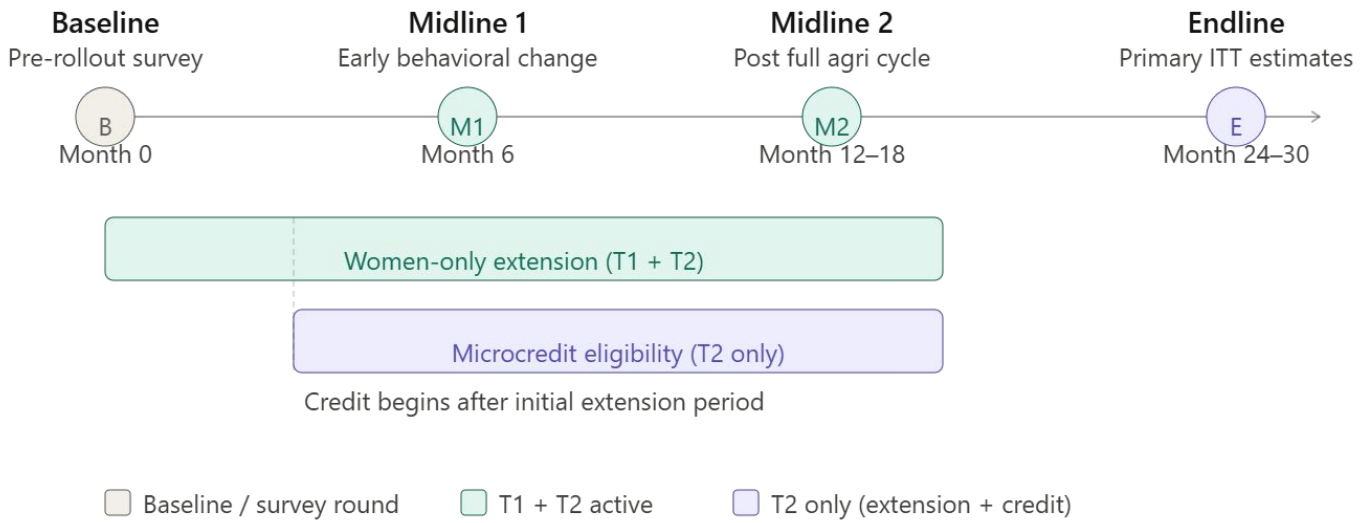
45 villages per treatment arm

~20 eligible women per villages (GHS median cluster size)

Study Design: Extension Details and Timeline

10 sessions over a growing season

- Two sessions per curriculum module
- 2 hours per session
- Bi-weekly: 20 weeks (one full growing season)
- Modules covered:
  - Crop Planning and Seasonal Decision-Making
  - Input Use
  - Basic Record-keeping
  - Post-harvest handling and Storage
  - Market-oriented decision-making
  - Negotiation skills



Data Collection & Outcome

Two Data Sources

GHS 2018-19

Existing national survey
Used for baseline parameters
(participation rate, ICC)

Our Panel Survey

Original data collection
4 rounds: Baseline, Midline 1 (~6mo),
Midline 2 (post-harvest), Endline (24-30mo)
Tracks intervention effects over time

Survey Instrument: 4 Modules

M1 Identification
Cascading lookup,
consent, hidden
treatment arm

M2 Demographics
Age, education,
land, HH roster
(denominator)

M3 Agri Labor
Days worked,
income, credit + consistency
checks

M4 Outcomes
LFPR, agri share,
decision-making
(pro-WEAI)

Primary Outcomes

Female LFPR

Women who worked in past 7 days (D1a)
÷ total adult women in household (B2b) × 100
ILO standard, 7-day recall

% Women in Agriculture

Women who worked on farm last season
÷ total adult women in household (B2b) × 100
Full-season recall

Baseline & Power Analysis

From GHS 2018-19 | N = 5,394 | 304 EAs | Rural adult women, age 15+

32.8%
Participation
past 7 days

0.228
ICC
2x literature avg

45.8 vs 25.1%
South vs North
20.7pp gap

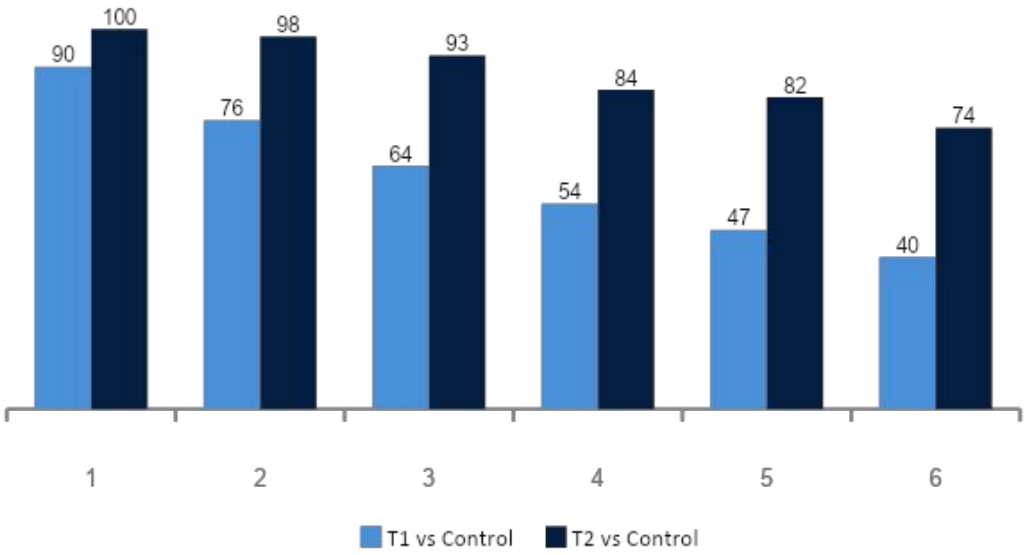
Simulation: 1,000 replications, clustered SEs

J/arm	Villages	HHs	T1 vs C (10pp)	T2 vs C (15pp)	T2 vs T1 (5pp)
15	45	900	.245	.433	.091
35	105	2,100	.432	.742	.156
45	**135**	**2,700**	**.486**	**.819**	**.152**
80	240	4,800	.712	.965	.241
100	300	6,000	.827	.992	.292

Recommended: 45 villages/arm (2,700 HHs). T2 vs C: 82% power. T1 vs C: underpowered. T2 vs T1: not feasible.

Sensitivity & Robustness

Power is highly sensitive to ICC (% at J=45)



Other findings

Attrition (20%)

T2 vs C power drops only from 81.4% to 80.0%. Robust to moderate dropout.

MDE threshold

At 45 villages/arm, we can reliably detect effects of 15pp or larger.

Regional heterogeneity

Cannot distinguish N vs S reatment effects at this sample. Exploratory only.

What We Cannot Answer

Microcredit marginal effect

The expected difference between extension+microcredit and extension-only is about 5 percentage points. At our ICC, this requires hundreds of villages per arm. Our study can assess the full intervention vs. status quo, but cannot isolate the credit component.

Regional heterogeneity

We pre-specify a test of whether treatment effects differ between North and South. However, at 45 villages per arm, the study has limited power for this comparison. This analysis will be treated as exploratory, not confirmatory.

Baseline parameters are conservative

The 7-day recall window may underestimate full-season participation (32.8% vs. potentially higher). ICC of 0.228 is above typical literature values (0.05-0.15). Both will be updated after the baseline survey. Final sample size may be adjusted.

Thank You!

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